

Module 3 Lab 3: Textbook Problem 2.2

CS 82A - Introduction to Data Science

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GitHub: https://github.com/ramishatasfia5156-commits/cs82a-coursework/blob/main/module3/module3_lab3.ipynb

```
In [1]: import pandas as pd

df = pd.read_csv('bridges.csv')

print('Shape:', df.shape)
print()
print('Missing values per column:')
print(df.isna().sum())
```

Shape: (108, 13)

Missing values per column:

Id	0
river	0
location	1
erected	0
purpose	0
length	27
lanes	16
clear-g	2
t-or-d	6
material	2
span	16
rel-l	5
type	2

dtype: int64

Plausibility Assumption

Length is the only numeric field that describes each bridge. It will be assumed to be in feet. Since all bridges in this dataset cross one of Pittsburgh's rivers (A = Allegheny, M = Monongahela, O = Ohio, Y = Youghiogheny), we can first write down our two rules before checking any flagged rows:

1. **Plausible range (impossible?):** Any river crossing must be at least long enough to span the river; therefore, any length less than 500 ft would likely be impossible. Also, we are assuming no single crossing could be longer than 5,000 ft (just under 1 mile). Therefore, we treat values outside 500 to 5,000 ft as errors.

2. **IQR fence (extreme?):** Values that fall above $Q3 + 1.5 \times IQR$ or below $Q1 - 1.5 \times IQR$ would be considered unusual for this dataset. We check these values against the other attributes of the bridge before determining what should be done.

A value outside rule 1 gets corrected or removed. A value outside only rule 2 is kept unless its other attributes contradict it.

Summary Statistics

```
In [2]: df['length'].describe().to_frame().T.round(1)
```

```
Out[2]:
```

	count	mean	std	min	25%	50%	75%	max
length	81.0	1567.5	747.5	804.0	1000.0	1300.0	2000.0	4558.0

Flagged Rows

```
In [3]: LOW, HIGH = 500, 5000
s = df['length']
q1, q3 = s.quantile(0.25), s.quantile(0.75)
iqr = q3 - q1
low_fence, high_fence = q1 - 1.5 * iqr, q3 + 1.5 * iqr
print(f'Plausible range: [{LOW}, {HIGH}]   IQR fences: [{low_fence:.0f}, {high_fence:.0f}]')

outside_range = (s < LOW) | (s > HIGH)
outside_fence = (s < low_fence) | (s > high_fence)
print('Outside plausible range:', outside_range.sum())
print('Outside IQR fence:      ', outside_fence.sum())

cols = ['Id', 'river', 'erected', 'purpose', 'length', 'lanes', 'material', 'span', 'type']
df.loc[outside_range | outside_fence, cols]
```

Plausible range: [500, 5000] IQR fences: [-500, 3500]

Outside plausible range: 0

Outside IQR fence: 3

```
Out[3]:
```

	Id	river	erected	purpose	length	lanes	material	span	type
31	E34	O	1888	RR	4558.0	2.0	STEEL	LONG	SIMPLE-T
44	E46	A	1897	RR	4000.0	2.0	STEEL	LONG	SIMPLE-T
104	E91	O	1975	HIGHWAY	3756.0	6.0	STEEL	LONG	ARCH

```
In [4]: # Context: how do the flagged bridges compare to other LONG-span steel bridges?
long_steel = df[(df['span'] == 'LONG') & (df['material'] == 'STEEL')]
['length']
```

```
print('LONG-span steel bridges:', long_steel.count())
print('Median length:', long_steel.median())
print('Longest unflagged:', long_steel[long_steel <= high_fence].max())
```

LONG-span steel bridges: 28
Median length: 1671.0
Longest unflagged: 2822.0

Handling the Flags

There is no length that falls outside the assumed minimum and maximum values for length (500 to 5,000 ft). Therefore, there is no value in this column that would be considered impossible under this assumption. Three bridges have lengths greater than the IQR fence of 3,500 ft:

E34 (4,558 ft), E46 (4,000 ft), E91 (3,756 ft) → extreme, not errors.

Action: kept.

All three are steel bridges with a LONG span, and their other attributes support their unusually long lengths. E34 and E91 both cross the Ohio River, which forms where the Allegheny and Monongahela meet. Additionally, E91 is a six-lane highway arch bridge from 1975, making it one of the newest and largest bridges represented in the data. Nothing in the other attributes of these bridges contradicts their lengths; therefore, removing them would remove real information.

The effect these extreme values may have on pulling up the mean length is handled through smoothing (below) rather than deletion.

Noise found outside length: E107's type is NIL, which is a placeholder rather than an actual bridge type. Therefore, it is treated as missing.

Missing Values (Textbook Part A)

```
In [5]: # E100 and E109 are missing length, lanes, t-or-d, material, span and type:
# every attribute used in Part C except purpose. Too little left to impute
# from.
print(df.loc[df['Id'].isin(['E100', 'E109']), cols])
df = df[~df['Id'].isin(['E100', 'E109'])].copy()

# NIL is a placeholder, not a type
df['type'] = df['type'].replace('NIL', pd.NA)

# length: median of bridges with the same purpose (overall median if the
# group has none)
group_median = df.groupby('purpose')['length'].transform('median')
df['length'] =
df['length'].fillna(group_median).fillna(df['length'].median())
```

```
# lanes: most common lane count for the same purpose
group_mode = df.groupby('purpose')['lanes'].transform(lambda x:
x.mode().iloc[0] if x.notna().any() else 2)
df['lanes'] = df['lanes'].fillna(group_mode).astype(int)

# categorical columns: explicit UNKNOWN instead of a guessed category
for col in ['location', 'clear-g', 't-or-d', 'material', 'span', 'rel-l',
'type']:
    df[col] = df[col].astype(object).where(df[col].notna(), 'UNKNOWN')

print()
print('Shape:', df.shape)
print('Missing values remaining:', df.isna().sum().sum())
```

	Id	river	erected	purpose	length	lanes	material	span	type
106	E100	0	1982	HIGHWAY	NaN	NaN	NaN	NaN	NaN
107	E109	A	1986	HIGHWAY	NaN	NaN	NaN	NaN	NaN

Shape: (106, 13)

Missing values remaining: 0

Smoothing (Textbook Part B)

```
In [6]: # Smoothing by bin means: 3 equal-frequency bins, each value replaced by
its bin's mean
df['length_bin'] = pd.qcut(df['length'], q=3, labels=['short', 'medium',
'long'])
df['length_smoothed'] = df.groupby('length_bin', observed=True)
['length'].transform('mean').round(0)

df.groupby('length_bin', observed=True)['length'].agg(['count', 'min',
'max', 'mean']).round(0)
```

```
Out[6]:
```

	count	min	max	mean
length_bin				
short	40	804.0	1200.0	1021.0
medium	32	1240.0	1460.0	1338.0
long	34	1500.0	4558.0	2241.0

Relationships (Textbook Part C)

```
In [7]: print('i. Length vs. purpose')
pd.crosstab(df['purpose'], df['length_bin'])
```

i. Length vs. purpose

Out[7]: **length_bin** **short** **medium** **long**

purpose			
AQUEDUCT	4	0	0
HIGHWAY	28	18	23
RR	8	13	11
WALK	0	1	0

```
In [8]: print('ii. Lanes vs. material')
pd.crosstab(df['material'], df['lanes'])
```

ii. Lanes vs. material

Out[8]: **lanes** **1** **2** **4** **6**

material				
IRON	1	10	0	0
STEEL	0	53	22	4
WOOD	3	12	1	0

```
In [9]: print('iii. Span vs. lanes')
pd.crosstab(df['span'], df['lanes'])
```

iii. Span vs. lanes

Out[9]: **lanes** **1** **2** **4** **6**

span				
LONG	0	17	10	3
MEDIUM	0	40	12	1
SHORT	2	6	1	0
UNKNOWN	2	12	0	0

Cleaning Log

Change	Before	After	Reason
Screen length	81 values	0 outside 500-5,000 ft; 3 outside IQR fence	Assumption stated before screening
E34, E46, E91 length	4,558 / 4,000 / 3,756	Kept	Extreme but consistent with LONG-span steel bridges
E100, E109	108 rows	106 rows	Missing every attribute used in Part C

Change	Before	After	Reason
E107 type	NIL	UNKNOWN	Placeholder, not a bridge type
length missing	25 NaN	0	Median length for the same purpose
lanes missing	14 NaN	0	Most common lane count for the same purpose
Categorical missing	NaN	UNKNOWN	Avoids inventing categories that would distort Part C
length_bin, length_smoothed	—	2 new columns	Smoothing by bin means (Part B), bins used in Part C

AI Disclosure

AI Disclosure: Claude (Anthropic) was used to help explain concepts and draft initial code for this lab. All code was reviewed, retyped, and understood by me. I can explain why the plausibility rule is stated before the screening code, how the range rule and IQR fence flag long bridges, why the three flagged bridges were kept, and how smoothing by bin means and `pd.crosstab` prepare the data for Part C.